SLEEPSENSE COMPANION

SleepSense

Turning Nightly Sleep Data into Simple Scores, Trends, and Actionable Routines

Demo Day Presentation • CircadianX Team

INTRODUCTION

The Core Proposition

The Problem We Solve

People sleep poorly, don't understand why, and struggle to stay consistent with healthy sleep hygiene routines. Existing apps track data but fail to bridge the gap to sustainable behavior changes.

Our Unified Solution

SleepSense integrates **Tracking + Habit Systems + AI Explanations** into a single clean mobile companion experience.

Key Outcomes

- **Speed-to-Insight:** Bulletproof visual "last night summary" within seconds of waking up.
- **7-Day Micro Trends:** Clean, color-coded weekly visual analysis rather than isolated nights.
- **Direct Accountability:** Gamified challenges and lightweight social features build true discipline.

TALK TRACK (15-25 SECONDS)

"Most tracking apps just throw charts at you. SleepSense bridges the gap by translating complex audio tracking directly into structured habits and personalized weekly AI reports."

SYSTEM ARCHITECTURE

Technology Snapshot

Frontend & Storage

Platform: Native Android

UI Framework: Jetpack Compose with modern declarative layouts.

Architecture: MVVM with reactive state streams.

Local DB: SQLite powered by Room ORM for offline stability and instant page loading.

Sensors & Tracking

Recording: Phone microphone-based analysis running via a persistent `Foreground Service`.

Hardware Integrations: Synced BLE connections (ESP32 microcontrollers) mapped straight to the main dashboard.

Seeding: Built-in mock data generators to ensure dynamic, live-looking trends during live demos.

AI & Cloud Analytics

AI Layer: Dual-endpoint LLM assistant backend integration.

Interactive Chat: `/chat` endpoint utilizing formatted context payloads (scores, durations, steps).

Weekly Report: `/report` endpoint performing pattern discovery across 14-day matrices.

DEMO STRATEGY

Recommended Presentation Flow

1. **Onboarding Walkthrough:** Fast-track setup (goals, target bedtime)
2. **Dashboard Reveal:** Highlight hero score (82), trends, & quick actions
3. **Live Recording Modal:** Demonstrate pulsing UI & sound waveform
4. **AI Weekly Report:** The heavy-hitting "WOW" feature summary
5. **History & Routines:** Drills down into the sessions database & habits
6. **AI Chat & Challenges:** Active prompt interaction & goals commitments

Demo Philosophy

Structure your pitch to highlight immediate visual feedback early, followed by the deep intelligence (AI Weekly Report), and finally showing how habits sustain the app's retention ecosystem.

Use the **Settings** → **Seed Data** function prior to presentation to ensure all historical tabs are populated.

PRESENTER TIP

"Move smoothly from the physical tracking elements to the automated habits to show a complete feedback loop."

STEP-BY-STEP FEATURE WALKTHROUGH

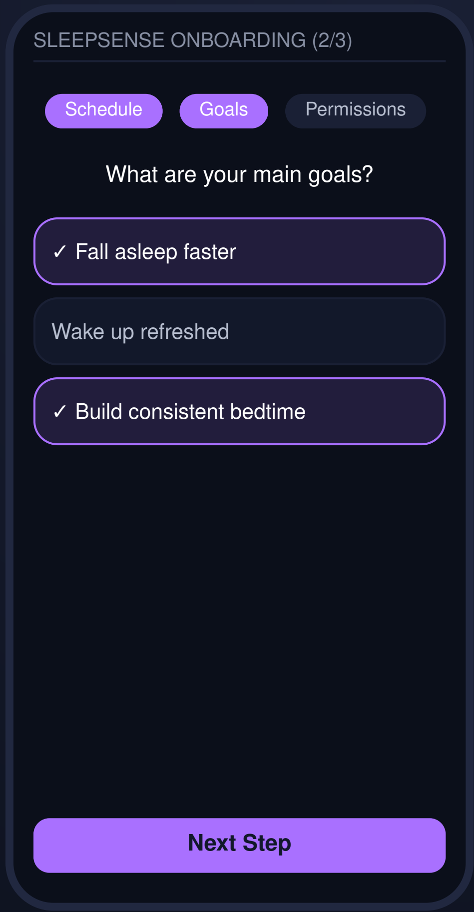
1. Onboarding Flow

Tailoring the Experience

Onboarding is lightweight but critical. It aligns tracking parameters around the user's specific target circadian rhythm.

- **Schedule Bed/Wake:** Establish baseline expectations for consistency metrics.
- **Goal Selection:** Users pick 1 to 5 goals (e.g., "Fall asleep faster", "Reduce waking up") that dynamically tune future AI summaries.
- **Permissions Bridge:** Seamless system dialog requests for system accessibility and mic access.

> **WHAT TO SAY**
>
> *"We immediately personalize tracking around your biological sleep targets. Permissions are structured elegantly to reduce initial drop-off."*



STEP-BY-STEP FEATURE WALKTHROUGH

2. Dashboard (Home Tab)

Last Night at a Glance

Designed for high-speed absorption immediately after waking up. It features three central pillars:

- **Hero Score Ring:** Single unified number (e.g. 82) with an adaptive qualitative color indicator ("Good").
- **Night Story:** Immediate human-readable translation ("Shorter than ideal, but next best night is reachable").
- **Night Timeline Strip:** Linear timeline indicating localized sleep disturbances (22:45 to 05:57).
- **Status HUD:** Synced device confirmations (e.g., "ESP32 synced").

> **WHAT TO SAY**
>
> *"The dashboard is built for speed. It tells you exactly what you need to know in under three seconds, using a color-graded score, a physical sleep timeline, and quick-access buttons."*

STEP-BY-STEP FEATURE WALKTHROUGH

3. Live Recording Screen

An Active, Calming UI

When active, the recording screen transforms into a distraction-free, low-light ambiance layout designed not to disrupt standard sleep prep.

- **Ambient Motion:** Gentle pulsing visual rings around a calm avatar center point.
- **Live Waveform:** Dynamic microphone input levels rendering real-time variations.
- **Instant Micro-Classification:** Audio thresholds automatically label states such as *Quiet*, *Breathing*, or *Snoring*.
- **Foreground Session Guard:** Protected via notification actions preventing standard memory teardown.

WHAT TO SAY

"The recording screen is designed specifically for dark bedside environments. It visualizes sound frequencies live, feeding our categorization engine while displaying simple diagnostic signals."

STEP-BY-STEP FEATURE WALKTHROUGH

4. AI Weekly Report (The WOW Feature)

True Insight Generation

Rather than standard stat lists, our LLM engine extracts narrative intelligence by cross-referencing multiple layers:

- **Multi-Factor Integration:** Analyzes previous 14 nights of sleep databases, physical steps count, and personal goals.
- **Clear Trend Markers:** Identifies comparative indicators (e.g. "+8% improvement vs last week").
- **Key Patterns Revealed:** Highlights relationships such as: *"Worst nights correlate with days under 5k steps."*
- **Honest Risk Assessment:** Clinical indicators (Sleep Apnea / fragmentation notes) accompanied by strict informational disclaimers.

WHAT TO SAY

"This is where our system excels. We synthesize qualitative sleep parameters, walking steps, and targeted goals to construct an structured summary detailing exactly how your day affects your night."

AI ANALYSIS • WEEKLY REPORT

APRIL 28 – MAY 3

78 ↑ 8%

Sleep score improved vs last week

Patterns Detected

- You sleep 42 min longer on weekends vs weekdays.
- Lowest scores match days with < 5k steps.

Risk Assessment

Low fragmentation risk. No symptoms of obstructive sleep apnea detected.

STEP-BY-STEP FEATURE WALKTHROUGH

5. History (Sleep History Tab)

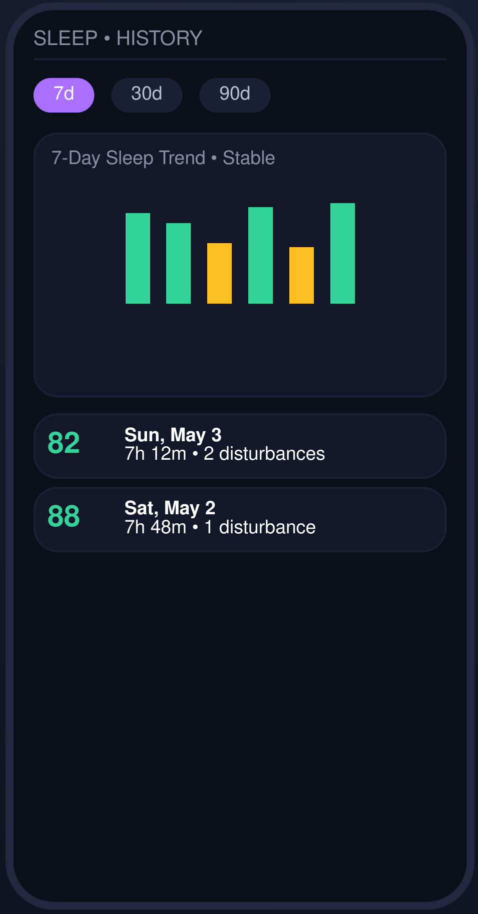
Validating Progress

History ensures long-term user validation. It is split into structural trend charts and micro session database details:

- **Flexible Range Chips:** Instant dynamic switching between 7-day, 30-day, and 90-day aggregations.
- **Visual Trend Bars:** High-impact, color-coded vertical graphs showing performance at a glance.
- **Historical Session Index:** Clean chronological cards with explicit metrics for each tracked night.
- **Bottom Sheet Disclosures:** Deep details showing exact bedtime, wake up time, and raw disturbance frequencies.

WHAT TO SAY

"The Sleep tab displays long-term trends. A user can track weekly consistency and tap any specific day to open a details sheet showing bedtime, wake patterns, and localized noises."



STEP-BY-STEP FEATURE WALKTHROUGH

6. Habits & Challenges Engine

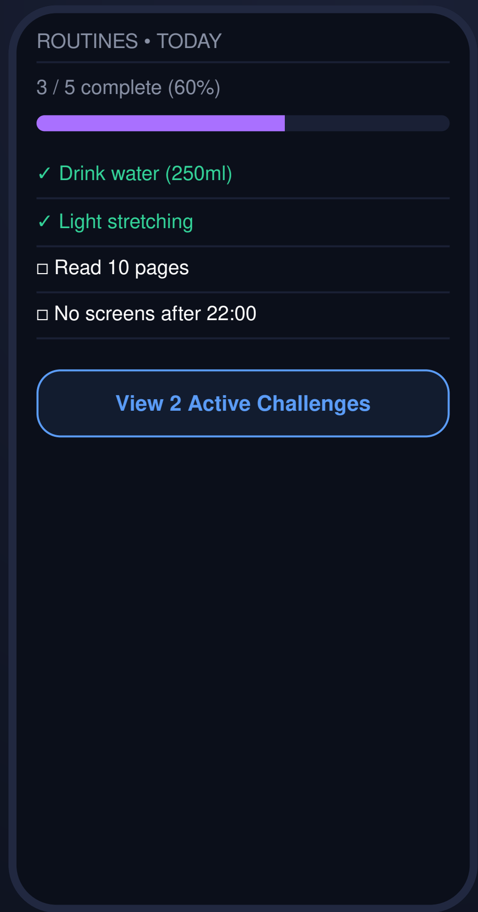
The Behavior Modification Loop

Tracking is passive. Habits are active. Our habit loops turn bedtime guidelines into structured behavioral goals:

- **Daily Checklists:** Separated into *Tonight* and *Morning* routines (e.g., "Brush teeth", "No screens after 22:00").
- **Progress Visualizers:** Real-time completion progress tracking (e.g., "3/5 complete • 60%").
- **Time-Bound Challenges:** Structured, multi-day challenges (e.g. "Sleep before midnight — at least 5/7 days").
- **Challenge Creator FAB:** Quick floating action buttons allow users to define custom durations and success markers.

WHAT TO SAY

"We help users change their behavior. By establishing concrete, customizable checklists, users turn tracking data into concrete habits, building discipline over time."



STEP-BY-STEP FEATURE WALKTHROUGH

7. Contextual AI Chat & Steps

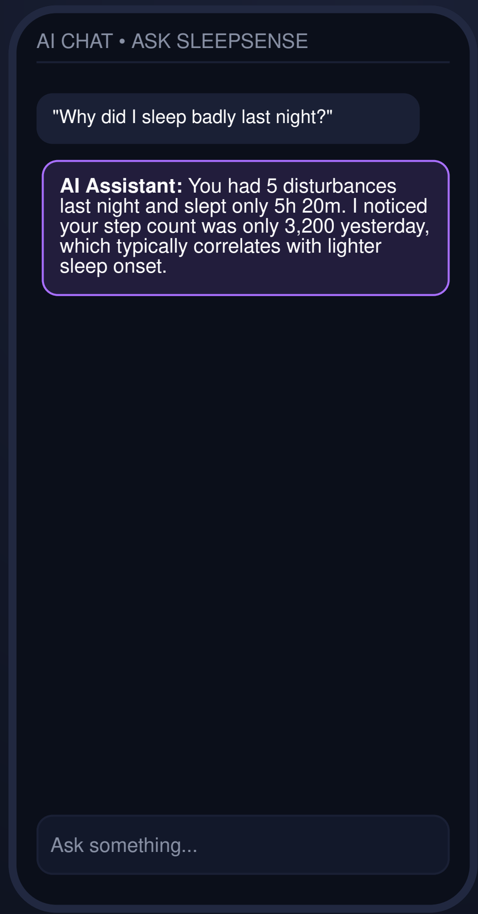
Personalized Conversational AI

Rather than generic, ungrounded recommendations, our AI chatbot utilizes active contextual signals:

- **Context Injectors:** Prompts pack deep data including sleep durations, disturbance trends, steps, and goals.
- **Actionable Responses:** The assistant refers directly to yesterday's low step counts or sleep disturbances.
- **Physical Activity Context:** Interactive step charts correlate daily activity levels with resulting deep sleep metrics.
- **Pre-Constructed Quick Prompts:** Accelerates engagement (e.g. "Why did I sleep badly last night?").

WHAT TO SAY

"The chat is context-aware. If you ask why you slept poorly, the AI pulls in your daily physical step count and last night's disturbances to provide an accurate, personalized response."



STEP-BY-STEP FEATURE WALKTHROUGH

8. Social, Security & Profile Settings

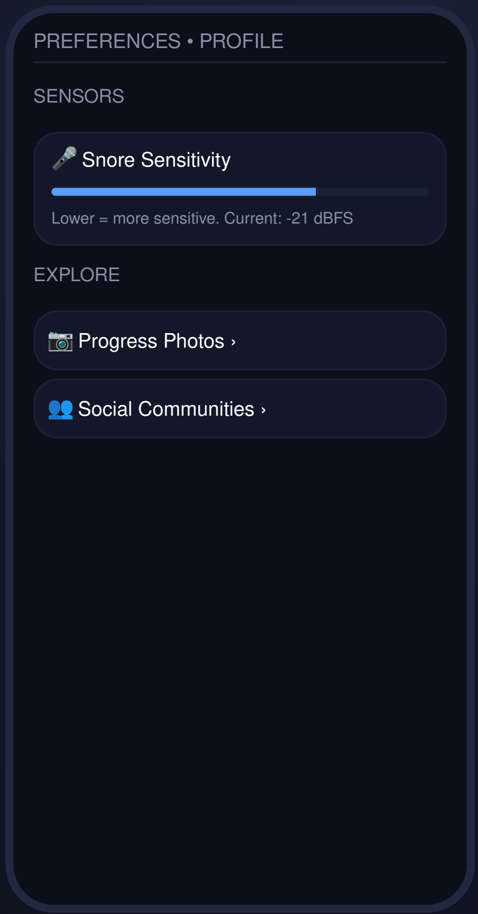
Security and Settings Architecture

Supporting features ensure account utility, security compliance, and community accountability:

- **Local Groups & Stories:** Encourages collaborative behavior through screen-free challenges and wake-up crews.
- **Progress Photos:** Encrypted visual tracking galleries (for physical well-being tracking), decrypted dynamically only when active.
- **Snore Thresholds:** Interactive decibel sensitivity adjustment controls (e.g., -21 dBFS).
- **Demo Utilities:** Seeding modules directly within settings for presentations.

WHAT TO SAY

"Our Profile tab handles advanced preferences. It configures snore microphone sensitivities, provides local encrypted storage for progress photos, and houses our demo data generator."



ENGINEERING INTEGRITY

Risks and Limitations

Key Considerations

- **Informational Only:** This application is not a clinical medical device. The AI Weekly Report is strictly educational.
- **Backend Dependencies:** The AI Chat assistant and Weekly Reports require reliable API server routing (e.g., `/chat` and `/report` connections).
- **Data Migration Path:** Active database cycles use destructive schema rebuilding during early prototype developments.

Mitigation Strategies

- **Offline Fallbacks:** The application runs independently utilizing local Room storage systems should server routes degrade.
- **Local Database Seeding:** Offline DB generators guarantee functional dashboards for demo environments without live tracking.

SleepSense

A Shipped, Presentation-Ready Sleep Companion App

Thank you! We are open to questions from the panel.

Kotlin | Jetpack Compose | Room DB | LLM Analytics